

Sri Lanka Institute of Information Technology



Lab Submission
Lab sheet No- 05

2026-Y2-S1-MLB-B6G2-07_Activity_Diagram
Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

Sequence Diagram.

1. Group Members & Module Allocations

ID: 2026-Y2-S1-MLB-B6G2-07

Project Topic : PawLife - Web Based Pet Care Management System

Student ID	Member Name	Assigned Module & Responsibility
IT25100618	Sanvidu S.D.N	System Administration & Business Intelligence Module
IT25101709	De Silva L.P.B	Digital Pet Profile & Health Record Module
IT25102521	Subasinghe R. A. G. I	Appointment Scheduling & Notification Module
IT25103607	Balasooriya B.A.H.N	Smart Inventory & Medical Supply Chain Management Module
IT25101534	Alahakoon A.M.B.U	Operational Centre & Staff Management Module
IT25103408	Warnakulasuriya G. A. D. T. S	Pet Grooming & Specialty Care Service Workflow Module

2. Individual Member Deliverables

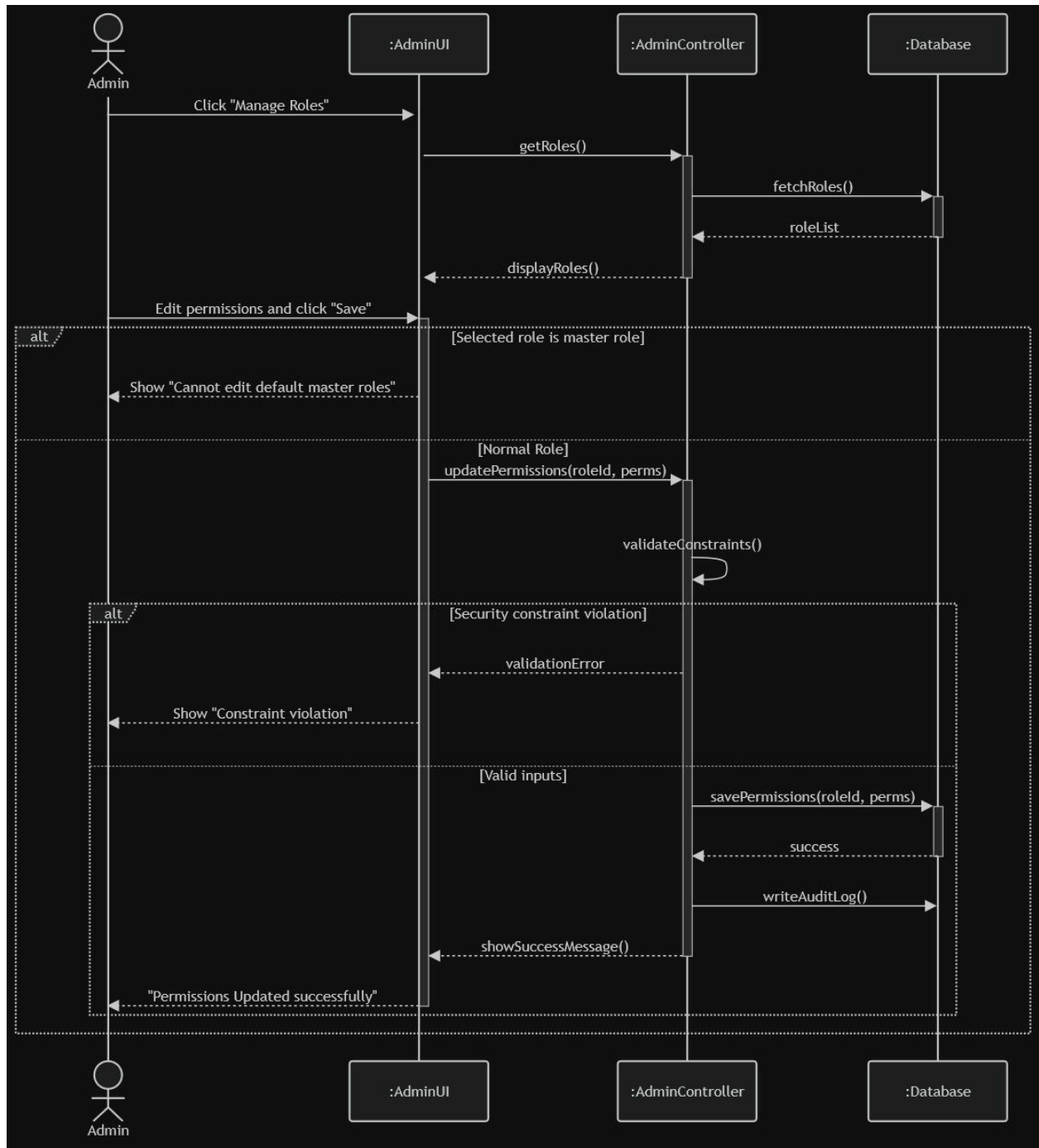
Member 01 - Sanvidu S.D.N (IT25100618)

Module: System Administration & Business Intelligence Module

Assigned Main Function : Configure Role-Based Access Control (RBAC) Permissions

Primary Actor	System Administrator
Pre-conditions	• Administrator is authenticated and logged into the PawLife admin dashboard with valid credentials. • System user roles and permission entity definitions exist in the central database.
Post-conditions	• Updated permission records are persisted to the database and reflected in active sessions. • An immutable audit log record is created for administrative accountability.
Main Success Scenario	1. System Administrator navigates to the RBAC Management panel in System Settings. 2. System retrieves and renders all system roles and the current permission matrix. 3. Administrator selects a target role to modify (e.g., Veterinary Officer, Centre Manager). 4. System displays granular permission checkboxes (Create, Read, Update, Delete) per module. 5. Administrator modifies privilege toggles and clicks "Save Changes". 6. System validates permission changes against system security constraints. 7. System commits the updated permission matrix to the database. 8. System concurrently triggers parallel updates: (a) refreshes the active session cache, and (b) writes a security audit log entry. 9. System displays a confirmation message: "Role permissions updated successfully." 10. Use case ends.
Extensions (Alternative Flows)	3.a. Selected role is a system-locked master role: ○ 3.a.1. System displays an alert: "Cannot edit default master roles."

	○ 3.a.2. Flow returns to Step 3. 6.a. Security constraint violation detected: ○ 6.a.1. System displays validation errors detailing conflicting privileges. ○ 6.a.2. Flow returns to Step 4.
--	---

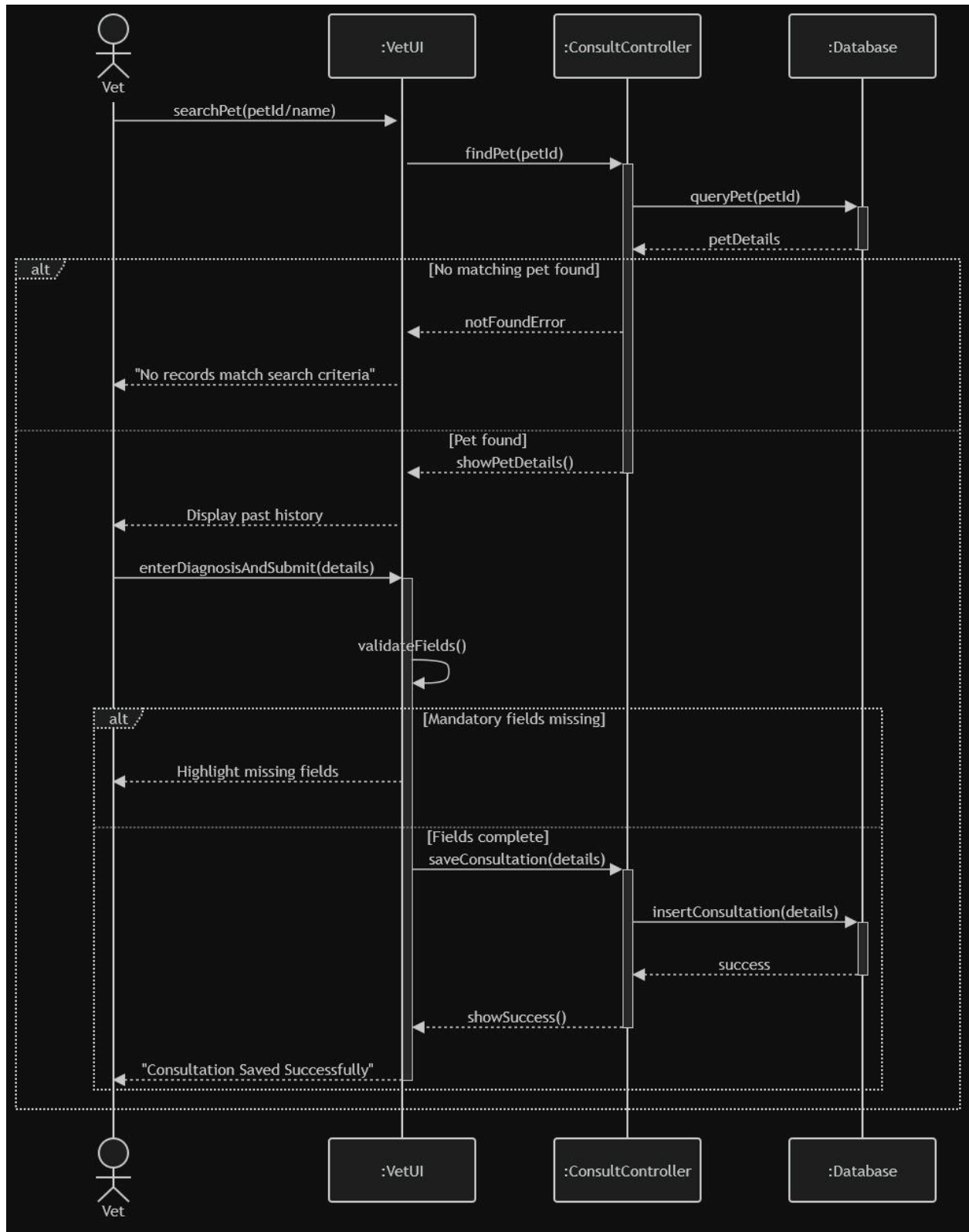


Member 02 - De Silva L.P.B (IT25101709)

Module: Digital Pet Profile & Health Record Module

Assigned Main Function : Search Pet Profile and Log Medical Consultation Notes

Primary Actor	Veterinary Officer
Pre-conditions	• Veterinary Officer is authenticated with valid clinical privileges. • Pet profile is registered in the PawLife central database.
Post-conditions	• Consultation notes, diagnoses, and treatments are appended to the pet's electronic health timeline. • Owner view is updated with current health status.
Main Success Scenario	1. Veterinary Officer enters search criteria (Pet ID, Pet Name, or Owner Phone Number). 2. System queries database and returns matching pet profiles. 3. Veterinary Officer selects target pet to view complete Electronic Health Record (EHR). 4. System displays past diagnoses, allergy warnings, and vaccination history. 5. Veterinary Officer clicks "New Consultation" and records symptoms, vitals, and diagnosis. 6. Veterinary Officer enters treatment notes and medication instructions. 7. Veterinary Officer submits the consultation record. 8. System validates that mandatory clinical fields are completed. 9. System commits the consultation to the pet's medical timeline. 10. System displays confirmation message and updates the record view. 11. Use case ends.
Extensions (Alternative Flows)	2.a. No matching pet found: ○ 2.a.1. System displays message: "No records match search criteria." ○ 2.a.2. Flow returns to Step 1. 8.a. Mandatory clinical fields empty: ○ 8.a.1. System highlights missing fields (e.g., Diagnosis required). ○ 8.a.2. Flow returns to Step 5.



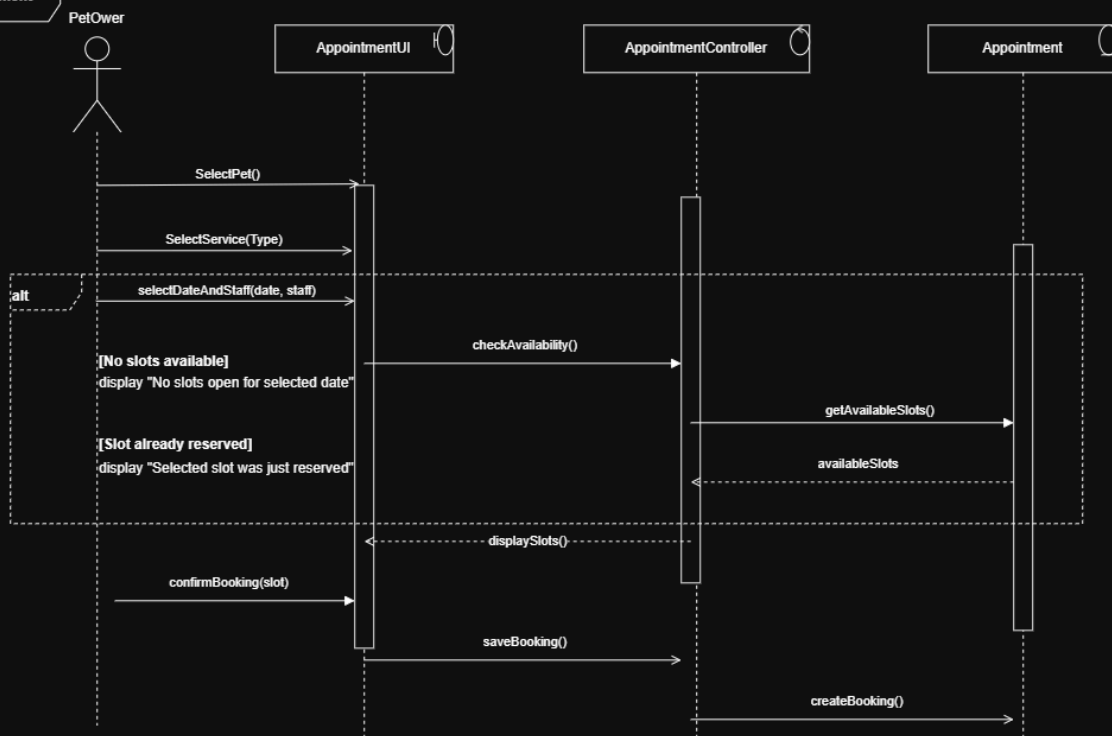
Member 03: Subasinghe R. A. G. I (IT25102521)

Module: Appointment Scheduling & Notification Module

Assigned Main Function : Book Veterinary or Grooming Appointment Online

Primary Actor	Pet Owner
Pre-conditions	• Pet Owner is logged into the PawLife portal. • At least one pet profile is registered under the owner account.
Post-conditions	• Appointment slot is locked and confirmed in the clinic schedule. • Confirmation notifications are sent to the owner and clinic schedule is synced.
Main Success Scenario	1. Pet Owner navigates to the Book Appointment section and selects registered pet. 2. Owner selects service category (Veterinary Care or Grooming). 3. Owner selects preferred date and practitioner. 4. System queries schedule in real time and displays open time slots. 5. Owner selects an available time slot and clicks "Confirm Booking". 6. System locks slot and writes booking record to database. 7. System concurrently triggers parallel actions: (a) sends SMS/Email booking confirmation, (b) syncs staff duty calendar. 8. System displays booking confirmation summary with appointment ID. 9. Use case ends.
Extensions (Alternative Flows)	4.a. No slots available on selected date: ○ 4.a.1. System displays message: "No slots open for selected date." ○ 4.a.2. Flow returns to Step 3 to choose alternate date. 6.a. Concurrent booking conflict detected: ○ 6.a.1. System alerts: "Selected slot was just reserved by another user." ○ 6.a.2. Flow returns to Step 4 to re-render remaining slots.

Book Pet Care
Appointment
Online

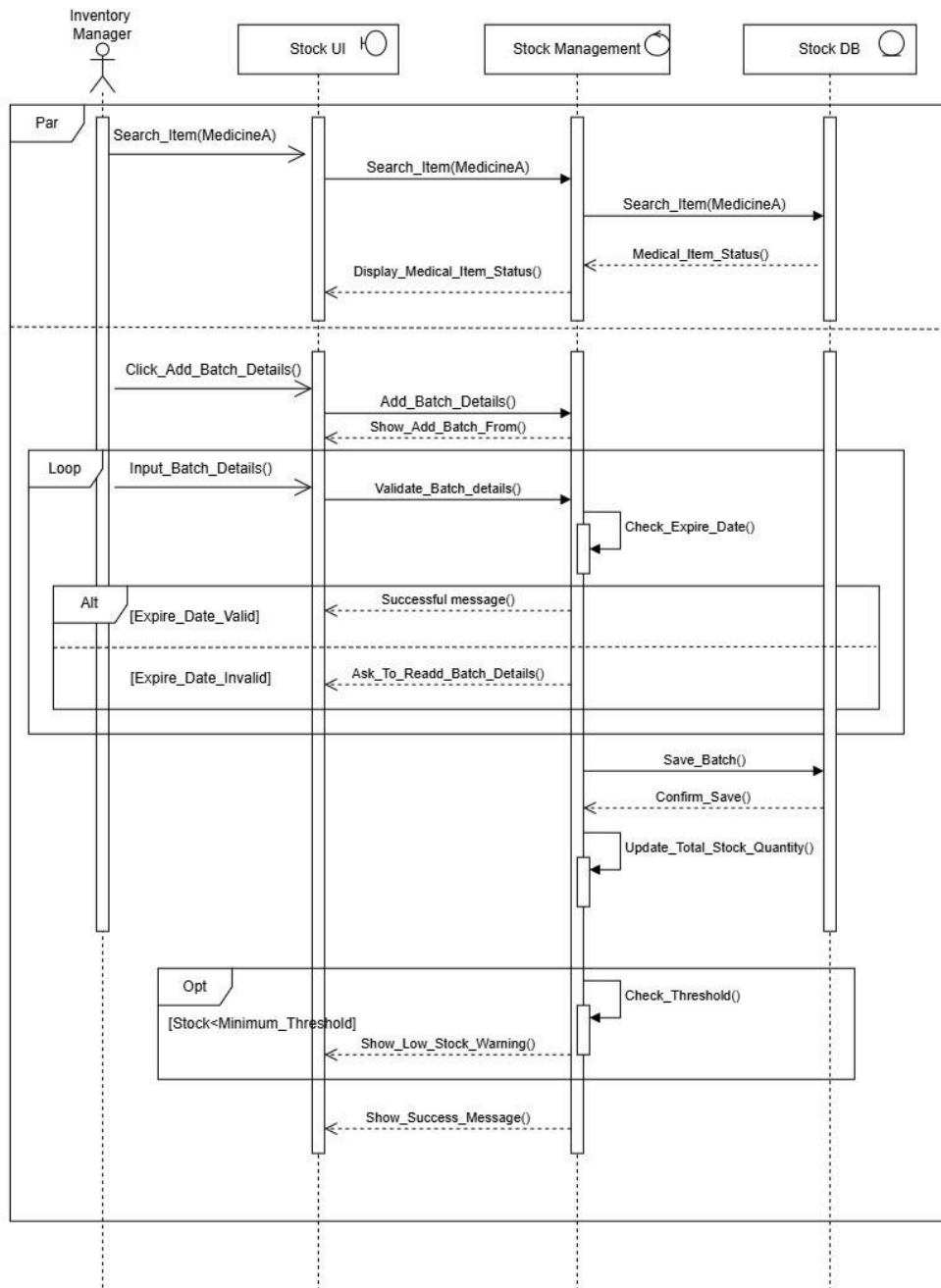


Member 04: Balasooriya B.A.H.N (IT25103607)

Module: Smart Inventory & Medical Supply Chain Management Module

Assigned Main Function : Track Medicine Stock Levels with Batch and Expiry Dates

Primary Actor	Inventory Manager
Pre-conditions	• Inventory Manager is authenticated with inventory permissions. • Medical supply item exists in master item catalog.
Post-conditions	• Batch details and quantities are updated in stock repository. • Low-stock and expiration threshold triggers are evaluated.
Main Success Scenario	1. Inventory Manager navigates to Stock Management panel and searches item by Code or Batch. 2. System displays item stock details and active batch records. 3. Manager clicks "Add New Batch" and enters received quantity, batch number, and expiry date. 4. System validates that expiry date is in the future and input format is correct. 5. System persists batch record and updates total available stock balance. 6. System evaluates if total stock is below reorder threshold. 7. System updates inventory dashboard and displays success summary. 8. Use case ends.
Extensions (Alternative Flows)	4.a. Expiry date is invalid or in past: ○ 4.a.1. System displays validation alert: "Expiry date must be in the future." ○ 4.a.2. Flow returns to Step 3. 6.a. Stock falls below low-stock threshold: ○ 6.a.1. System triggers automated Low-Stock Alert on manager dashboard. ○ 6.a.2. Flow proceeds to Step 7.



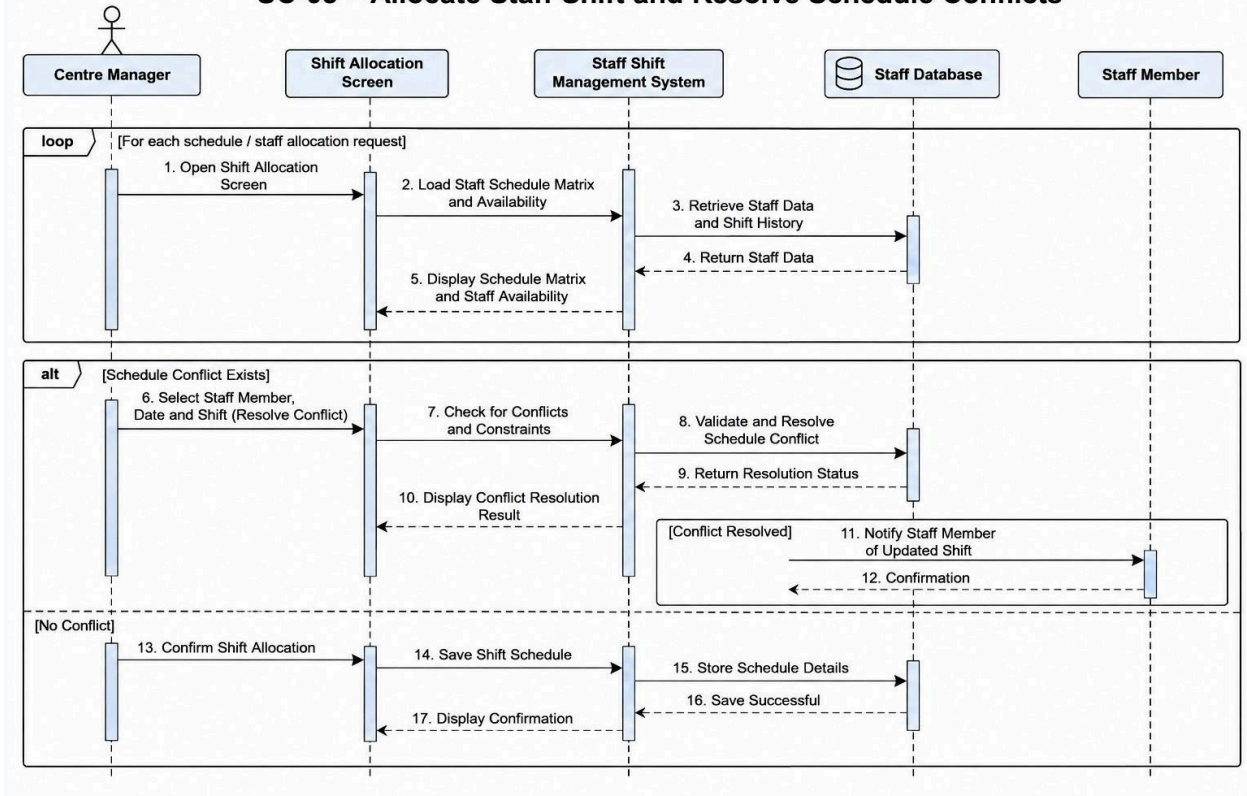
Member 05: Alahakoon A.M.B.U (IT25101534)

Module: Operational Centre & Staff Management Module

Assigned Main Function : Monitor Daily Clinic Dashboard and Allocate Staff Shifts

Primary Actor	Pet Care Centre Manager
Pre-conditions	• Centre Manager is logged into the management portal. • Active staff accounts are available in the system.
Post-conditions	• Staff shift assignment is committed to the operational roster. • Shift schedule notification is dispatched to the staff member.
Main Success Scenario	1. Centre Manager navigates to the Staff Shift Allocation screen. 2. System loads weekly schedule matrix and staff availability. 3. Manager selects staff member, date, and designated shift slot. 4. Manager clicks "Assign Shift". 5. System validates schedule conflicts and working hour limits. 6. System confirms no conflict exists. 7. System persists shift assignment to the database. 8. System concurrently triggers parallel tasks: (a) updates live operational roster, (b) dispatches shift alert to staff member. 9. System displays confirmation message: "Shift assigned successfully." 10. Use case ends.
Extensions (Alternative Flows)	5.a. Shift conflict or maximum hours exceeded: ○ 5.a.1. System displays warning modal detailing the overlapping assignment. ○ 5.a.2. Flow returns to Step 3 for re-assignment.

UC-05 – Allocate Staff Shift and Resolve Schedule Conflicts



Member 06: Warnakulasuriya G. A. D. T. S (IT25103408)

Module: Pet Grooming & Specialty Care Service Workflow Module

Assigned Main Function : Track Grooming Session Progress and Log Notes

Primary Actor	Grooming Specialist
Pre-conditions	• Grooming Specialist is logged into station terminal. • Pet has been checked in for an active grooming appointment.
Post-conditions	• Grooming status is updated and synchronized with Pet Owner portal. • Coat condition and observation notes are recorded.
Main Success Scenario	1. Groomer opens Active Grooming Queue on station terminal. 2. System displays checked-in pets waiting for grooming. 3. Groomer selects target pet and advances workflow status (Bathig - Styling - Ready). 4. Groomer inputs coat condition or behavioral observation notes. 5. Groomer clicks "Update Status". 6. System saves workflow state and observation notes to database. 7. System checks if the updated stage is "Ready for Pick-up". 8. System publishes live tracking status to Pet Owner Portal. 9. System displays confirmation and refreshes queue. 10. Use case ends.
Extensions (Alternative Flows)	7.a. Stage equals "Ready for Pick-up": ○ 7.a.1. System automatically sends automated SMS/Push notification alert to pet owner. ○ 7.a.2. Flow proceeds to Step 8.

